Prabhavit Gupta



AI Engineer: Generative AI, Data Pipelines & LLM Production Systems

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Summary

AI Engineer with 3 years of hands-on experience designing, building, and deploying production Generative AI applications, intelligent data pipelines, and scalable backend services. Proven track record developing and deploying production-grade AI systems grounded in live relational databases and third-party APIs, optimizing retrieval pipelines, and engineering automated evaluation and testing harnesses. Strong proficiency in Python, TypeScript, SQL, Docker, FastAPI, and data transformation architectures. Highly autonomous builder operating daily with AI-native tools (Claude Code) to move rapidly from requirements to robust production deployments. Based locally in Venice, Italy with permanent EU work authorization (EU Long Term Residence Permit) and immediate availability for Padua / Milan hybrid roles.

Technical Expertise

- **Generative AI & NLP Solutions:** LLM application engineering, RAG architectures, Semantic search, Prompt optimization, Tool/function calling, Structured JSON generation, Guardrails & fallback logic.
- **Data Pipelines & Backend:** Python, FastAPI, Node.js, TypeScript, RESTful API design, Webhooks, Automated data ingestion, Schema validation, Asynchronous task processing.
- **Data Stores & Infrastructure:** PostgreSQL, Supabase, Relational database modeling, SQL indexing, Docker, Containerized deployments, Cloud services (AWS/GCP), CI/CD pipelines, Git.
- **Testing & Observability:** End-to-end automated testing with Playwright, Trace analysis, Production transcript evaluation, Error logging, Latency profiling & token budgeting.
- **AI-Native Development:** Daily default power user of Claude Code and modern AI agent tooling for high-velocity software engineering, automated refactoring, and rapid prototyping.
- **Languages & Authorization:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long Term Residence Permit (Permanent unrestricted work authorization in Italy without visa sponsorship).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Built and operated the production AI customer support agent on a platform serving 4,500+ active users, handling live inquiries on pricing, inventory, and bookings.
- Grounded agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus), engineering strict tool-call schemas and fallback logic that escalated to humans on low confidence.
- Evaluated model performance and failure modes against real production transcripts rather than synthetic benchmarks, systematically eliminating hallucinations in inventory and pricing answers.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window usage to reduce operational cost while maintaining rapid response times.
- Deployed automated end-to-end test suites using Playwright, verifying API contracts and continuous integration stability across customer booking journeys.
- Spearheaded platform migration from a legacy stack to a modern Next.js and Laravel architecture, scaling capacity from 2,000 to 4,500+ users.
- Built automated data ingestion pipelines that parsed and validated unstructured partner listings, scaling platform inventory to 900+ active locations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Architected web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic.
- Implemented automated validation pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring data consistency.
- Integrated CI/CD workflows and automated regression checks to accelerate feature delivery.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed bespoke AI workflows, automated web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, repetitive business operations into automated software solutions with structured logging and error recovery.
- Engineered webhook listeners and real-time synchronization pipelines across CRM and operational tools.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous foundation in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.